

INTELLIGENT AGRICULTURE

Version 3.0

Turkey's Agricultural Digital Transformation

The Opportunity

- 3 million farms in Turkey
- 23 million hectares under management
- ~9 million rural population
- 60-80 Billion TL annual production value
- TAM: 4.5-5.75 Billion TL (farmers willing to spend 500 TL/year)

Four Critical Gaps

Gap 1: Production Planning — 85% of farmers use intuition ($\pm 25\%$ yield variance)

Gap 2: Expert Access — $< 10\%$ use paid advisory (geographically isolated)

Gap 3: Market Transparency — No farm-to-table visibility (commodity pricing)

Gap 4: B2B Inefficiency — 40% farmer income lost to wholesaler middlemen

Financial Impact Per Farmer

Current: Yield variance $\pm 25\%$, Cost opacity, 12.5% income loss

With IA Platform:

- +20% yield improvement
- -15% cost reduction
- +20-30% premium via transparency
- Estimated farmer income gain: +15-20% per season (~150K TL)

The Solution: 5-Layer Architecture

Layer 1: Farmer Portal (Web, Mobile, Voice) — Offline-capable

Layer 2: Six Integrated Modules (FMIS) — Production, Monitor, Livestock, Equipment, Helpdesk, Market

Layer 3: Service Integration — Expert network, payments, state systems

Layer 4: Data Layer — Cloud databases, APIs, analytics

Layer 5: ICT Backbone — Information (satellite, AI) + Communication (mobile, SMS, offline)

Multi-Modal Data Ingestion

Platform accepts data naturally farmers already use:

- Voice (STT + NLP) — "Irrigate field tomorrow"
- Image (OCR + CV) — Invoice photo → Expense; Disease photo → AI diagnosis
- Video (Action Recognition) — 30-sec field walkthrough → Crop status
- SMS/WhatsApp — Command execution
- Sensors (IoT + Satellite) — Auto-feed, no manual entry
- AR (Real-time Diagnosis) — Point phone at crop → Treatment overlay

Six Core Modules (FMIS)

FMIS.Agro — 90-day production planning, cost accounting, daily discipline

FMIS.Monitor — Satellite + IoT crop health, AI disease detection (72-hr early warning)

FMIS.Livestock — Wearable sensors, AI veterinary diagnosis

FMIS.Equipment — P2P machinery rental marketplace (3 tiered options)

FMIS.Helpdesk — 24/7 AI chatbot + expert escalation

FMIS.Market-Access — B2B direct sourcing + B2C QR traceability

Market Positioning

NO integrated ag-tech competitor in Turkey

First-mover with:

- Production planning + expert network + market access (integrated)
- Designed for seamless State data integration
- Local expert network (Dr. Koçak, 5,000+ agronomist ecosystem)
- "Self-sustaining" model (VC / Sponsor - independent)

Revenue Model (5 Streams + Sponsorships)

Platform Revenue

- Premium subscription: 500 TL/month
- Expert marketplace: 10-15% commission
- B2B sourcing: 5% commission
- B2C traceability: 8% commission
- Equipment P2P: 15% commission

Sponsorship Revenue

- Banks (payment integration)
- Food companies (sourcing access)
- Input suppliers (marketplace presence)
- Insurance (premium targeting)

Y1: 850K → Y3: 14.1M TL

Financial Projections

Year 1: 2.7M TL revenue | Break-even Month 9

Year 3: 70.38M TL revenue | 91% operating margin

Year 5: 113M TL revenue | 89% operating margin

Unit Economics:

CAC: 1,500 TL (organic) → 20 TL (sponsorship-aided) = 75x improvement

LTV: 59K TL | Payback: 3 months | LTV/CAC: 40:1

Implementation Roadmap

MVP (M0-6)

- 20-50 pilot farmers
- Core modules: Agro + Monitor
- B2B pilot launch
- 30-50 expert onboarding

Investment: ~3M TL

Phase 1 (M6-12)

- Scale to 100-300 farmers
- Expert marketplace launch
- Livestock module
- 3+ food company partnerships

Series A: 2.5-3.5M TL

Data Accumulation Strategy (Y1-Y5)

Year 1-2: AI Baseline — Satellite + weather data (80% regional accuracy)

Year 2-3: Crowdsourced — 150K+ expert assessments (parcel-level)

Year 3-5: Proprietary Intelligence — $\pm 5\%$ accuracy, B2B data licensing (15-20M TL/year)

Result: Build Maschinenring's 60-year moat in 3-5 years via digitization

Defensible competitive advantage = racketeer entry barrier

Unit Economics & Margins

Gross Margin: 80-85% (SaaS + commission model)

Operating Margin Progression:

Year 1: ~8% (investment phase)

Year 3: 91% (scale + sponsorship leverage)

Year 5: 89% (mature platform)

Churn Target: <5% monthly (high engagement from expert network)

Why We Win

- ✓ First-mover in integrated ag-tech (no Turkish competitor)
- ✓ Defensible data moat (Y3+ $\pm 5\%$ accuracy predictions)
- ✓ Expert network (Dr. Koçak credibility, day-1 trust)
- ✓ Exceptional unit economics (CAC 20 TL with sponsorships)
- ✓ Sustainable financing (not VC-dependent; sponsorships + platform revenue)
- ✓ Mission-driven (farmers earn +30%, agriculture sustainable)

Funding Ask: 4.5M TL (~\$1.5M USD)

Product & Tech

Platform MVP: 1.2M TL

Mobile app: 150K

AI/ML pipeline: 100K

Satellite integration: 25K

Infrastructure: 80K

Operations & Go-to-Market

Team (5-6 engineers, ops): 1.5M TL

Marketing & acquisition: 0.8M

Infrastructure & hosting: 0.5M

Buffer & contingency: 0.5M

The Team

Dr. Ahmet Koçak — Leading agronomist

→ 4.7/5 expert rating, 5,000+ expert network, day-1 credibility

[CEO/Founder] — Platform vision, technology leadership

[CTO] — AI/ML, satellite data, blockchain expertise

[COO] — Agricultural operations, farmer relationships

Hiring: Senior engineers, customer success, partnerships by Month 2

State Data Integration

Architecture designed for seamless State system integration:

- LPIS field registry data
- Satellite data (Copernicus, free tier)
- TARSIM historical yield data
- TSMS weather data

IA = Operations layer that transforms State data into farmer outcomes

Differentiation: Translation, experts, market access, revenue model

Risk Mitigation

Technology Risk — Copernicus validated globally; AI model v1 tested in MVP (20 farms)

Market Risk — MVP with Dr. Koçak endorsement (trusted voice); incentive-aligned

Competitive Risk — 3-year head start, expert network defensible, data moat (Y3+)

Regulatory Risk — KVKK audit in MVP; ISO 27001 by Phase 1

Financial Risk — Platform profitable without sponsorships (slower growth)

Next Steps (Q3-Q4 2026)

- ✓ Pilot launch: 20 farmers (proof of concept)
- ✓ MVP modules: Agro + Monitor + Market-Access beta
- ✓ Expert network: 30+ agronomists onboarded
- ✓ B2B pilot: 1-2 food companies testing

Success metrics: 20+ active users, <10% churn, +15% yield improvement, 3 companies interested

The Ask

4.5M TL Seed Capital | Turkey's Agricultural Intelligence
Revolution